LO-World

Unified BEV Latent Modeling for Joint 4D Occupancy and LiDAR Forecasting

Joint 4D semantic occupancy & LiDAR forecasting in one BEV latent

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Shared latent co-predicts occupancy & points; both improve

LO-World fuses semantic occupancy and LiDAR into **a single BEV latent** via hybrid "range image → voxel → BEV" encoding, then a flow-matching generator (context computed once + 4-layer shallow flow network) forecasts both modalities 3 s ahead with **4 sampling steps**.

57.46%

Occupancy Avg IoU

LO-World-T (occupancy + LiDAR input)

+10.9% relative to GenieDrive (51.80%)

0.58 m^2

LiDAR Avg Chamfer Distance

LO-World-TBM (+ boxes & map)

−30.1% relative to LaGen (0.83 m^2)

25.31

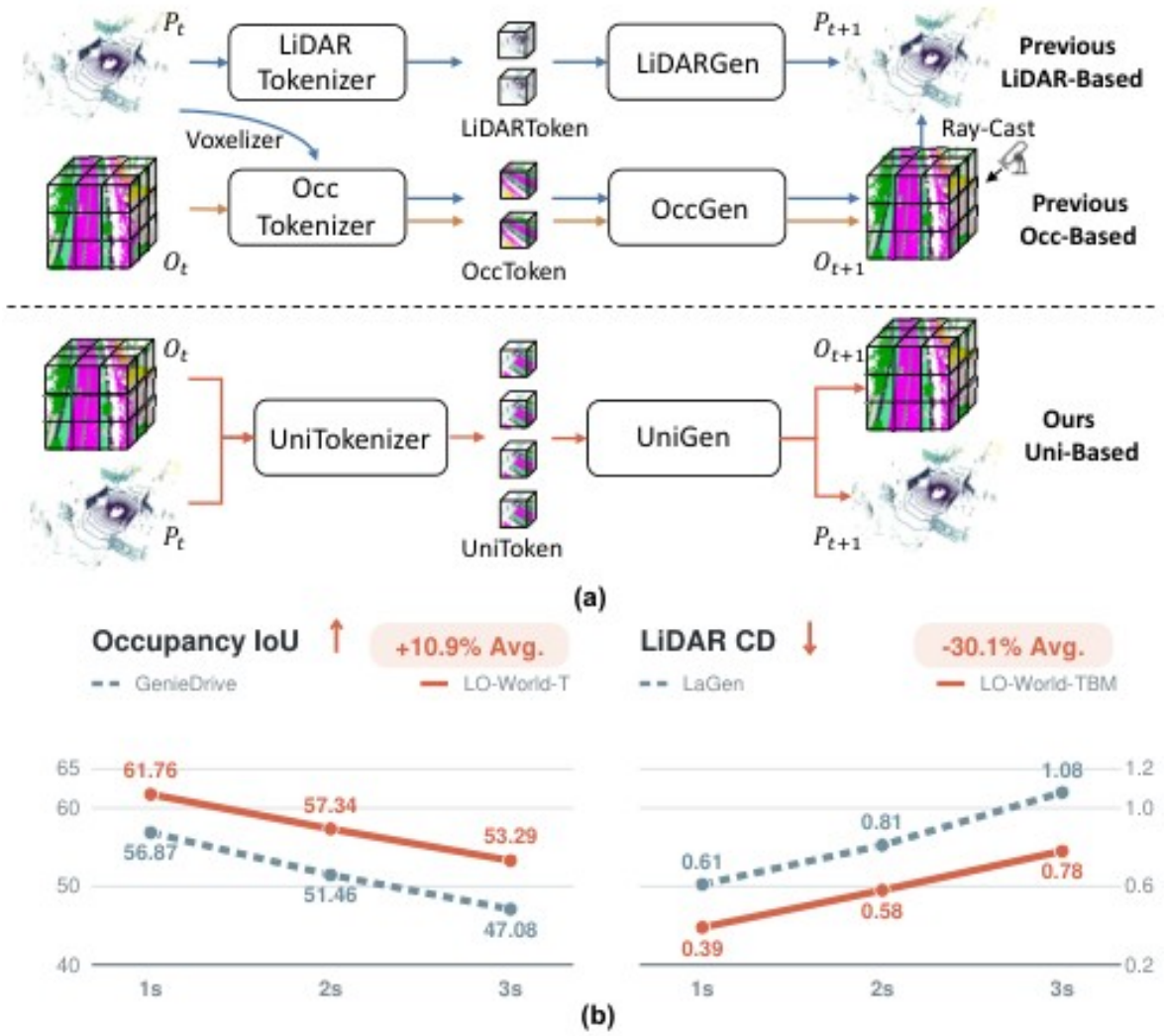
FPS-E2E (single RTX 4090)

Incl. decoding 6 future occupancy + LiDAR frames

OpenDWM-DiT: only 0.10

Note: with identical input (occupancy only), LO-World-T* reaches 53.47% IoU vs. GenieDrive's 51.80% (≈ +3.2%); joint modeling of both modalities raises it to 57.46%.

Siloed paradigms: no mutual constraint between occupancy and LiDAR



Sem. occ.: dense, blurry edges

0.4 m voxelization blurs out fine structures

LiDAR points: precise but sparse

Metric accuracy, no semantics, sparse at range

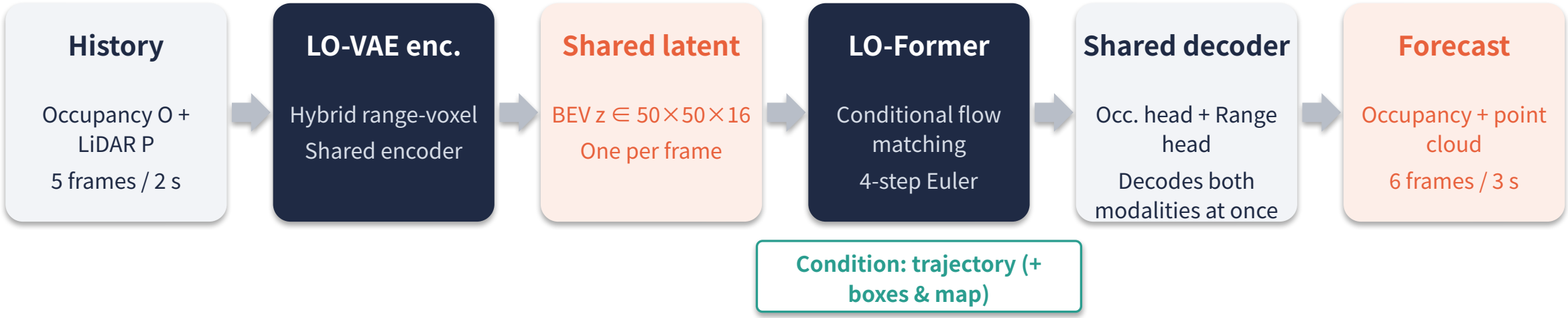
Limitations of prior paradigms

Separate: 2 passes, no cross-modal link

Cascade (UniScene): no LiDAR→occ. loss

Key question: can a single generative process let dense semantics and fine geometry regularize each other?

Two stages: LO-VAE unified tokenizer, LO-Former latent forecasting



Objective: $p(P_{\tau'}, O_{\tau'} \mid P_{\tau}, O_{\tau}, C_{\tau'})$ one joint conditional distribution over future points P and occupancy O; C = future trajectory / boxes / map

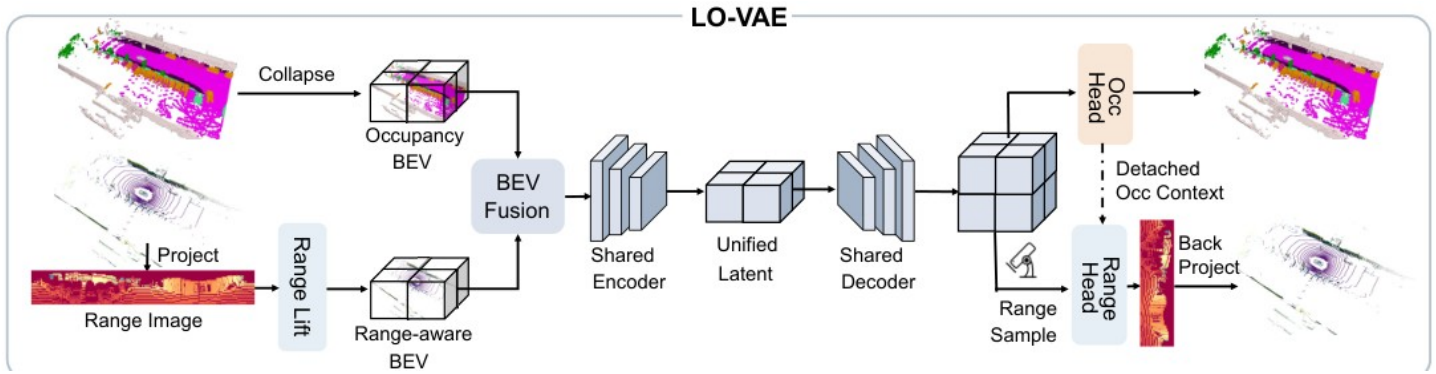
Stage 1: train LO-VAE (40 epochs)

Reconstruct paired occupancy and LiDAR; loss = occupancy loss + range loss + KL (10^{-6})

Stage 2: freeze VAE, train LO-Former

2 frames, then 6; for 6 s forecasts, fine-tune on 12 frames, autoregressive in two chunks

LO-VAE: hybrid range-voxel encoding aligns both modalities at low cost



LO-VAE architecture (top of Fig. 2 in the paper)

Ablation (Table 6): plain voxelization gives CD 0.70 → 0.92, FPD 5.16 → 75.23; high-res voxels reach CD 0.75 but drop speed 26.3 → 6.7 FPS.

① Project to range image

32×1024; per pixel [log depth, valid bit], native scan lines kept

② Voxel pooling → BEV

Unproject to 0.4 m voxels, avg-pool, collapse Z; only 2×16 channels

③ Shared encoder

Per-modality stem → concat → ResNet → VAE latent 50×50×16

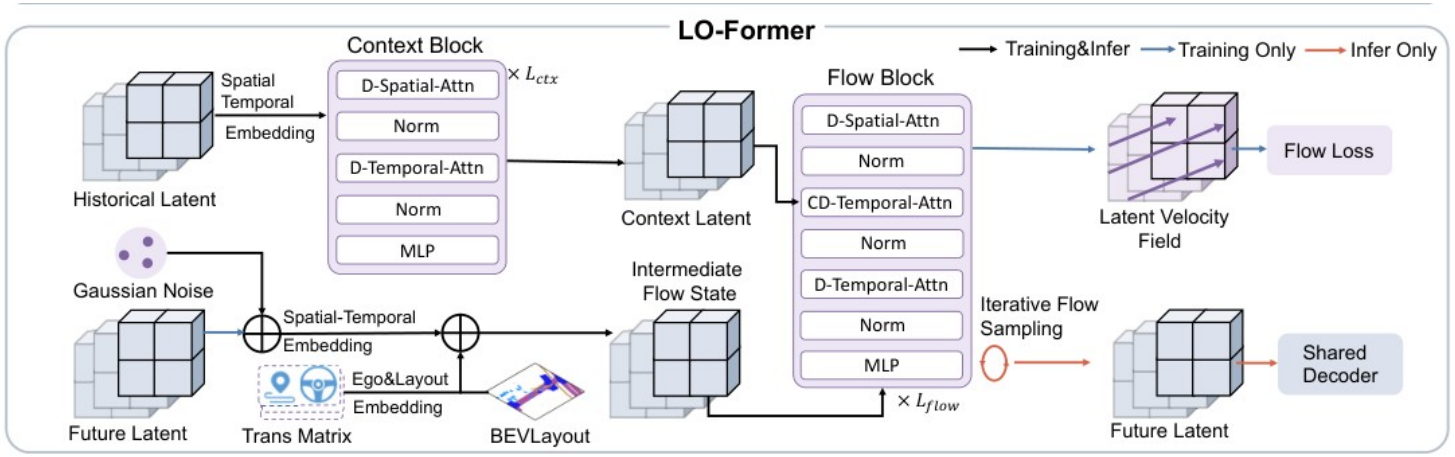
④ Dual-head decoding

Occ. head: Focal + Lovasz

Range head: 192 depth bins/ray + empty class

Occupancy probs as stop-grad prior

LO-Former: history encoded once; only a 4-layer shallow net per step



LO-Former architecture (Fig. 2 bottom; blue arrows: training only, orange: inference only)

Training: $x_\sigma = (1-\sigma)x_0 + \sigma\varepsilon$; regress velocity $v = \varepsilon - x_0$

Inference: Gaussian noise → 4 Euler steps → 6 future latents → shared decoder

8 layers

Context block • run once

DSA + DTA aggregate spatio-temporal history; output H cached

4 layers

Flow net • every step

DSA + CDTA + DTA; CDTA sparsely samples from H

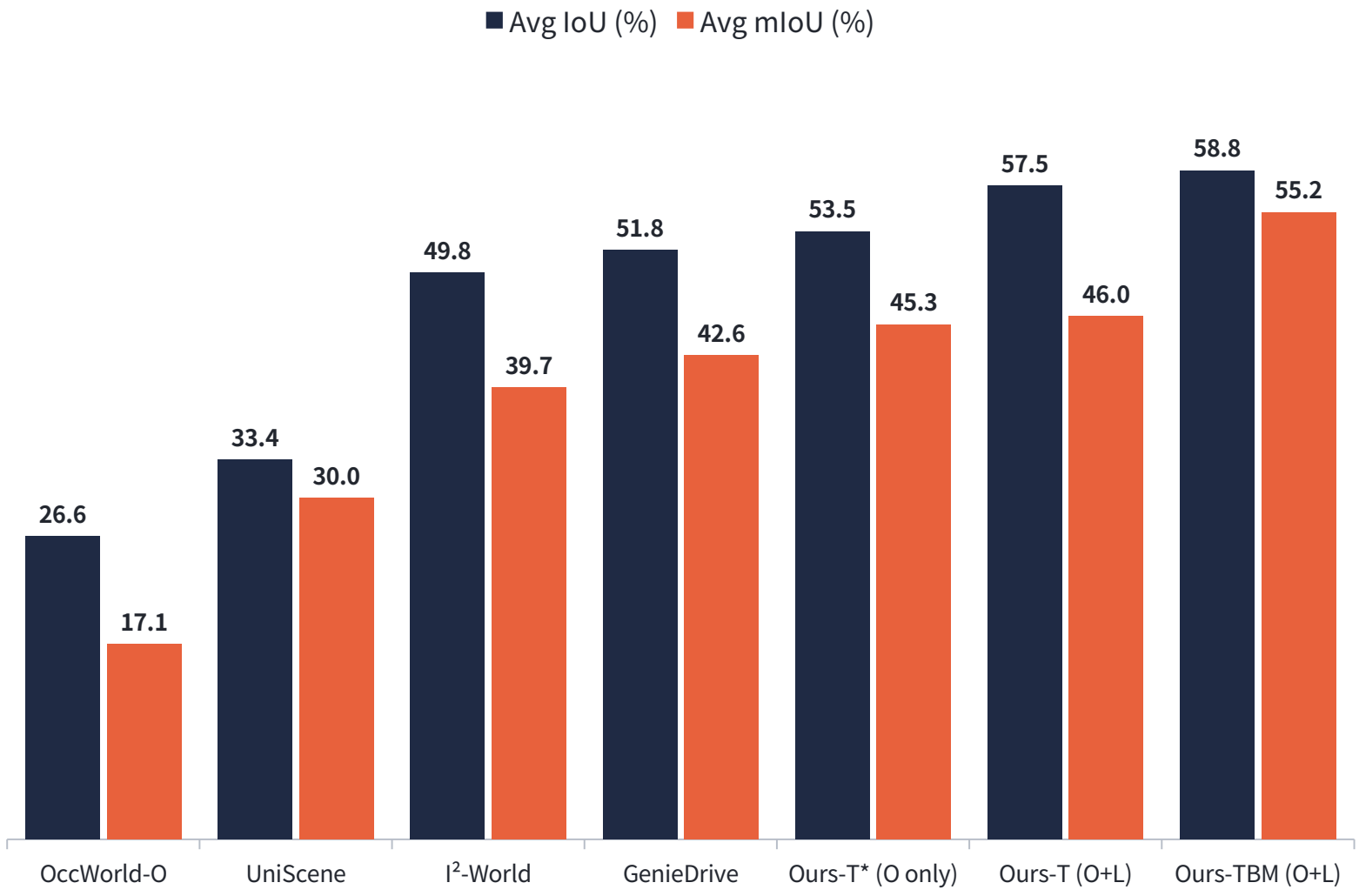
4 steps

Euler sampling

Hidden dim 512, 8 heads; 43.2M params

Source: Fig. 2 bottom (LO-Former), Sec. 3.2, Table A2

Occupancy: IoU hits 57.46% with LiDAR, beating all baselines



T* leads with one modality

IoU 53.47 vs GenieDrive 51.80

mIoU 45.28 vs 42.59

+ LiDAR input

IoU 53.47 → 57.46

Model FPS still 45.2

+ Boxes & map (TBM)

mIoU 46.01 → 55.15 (+19.9% rel.)

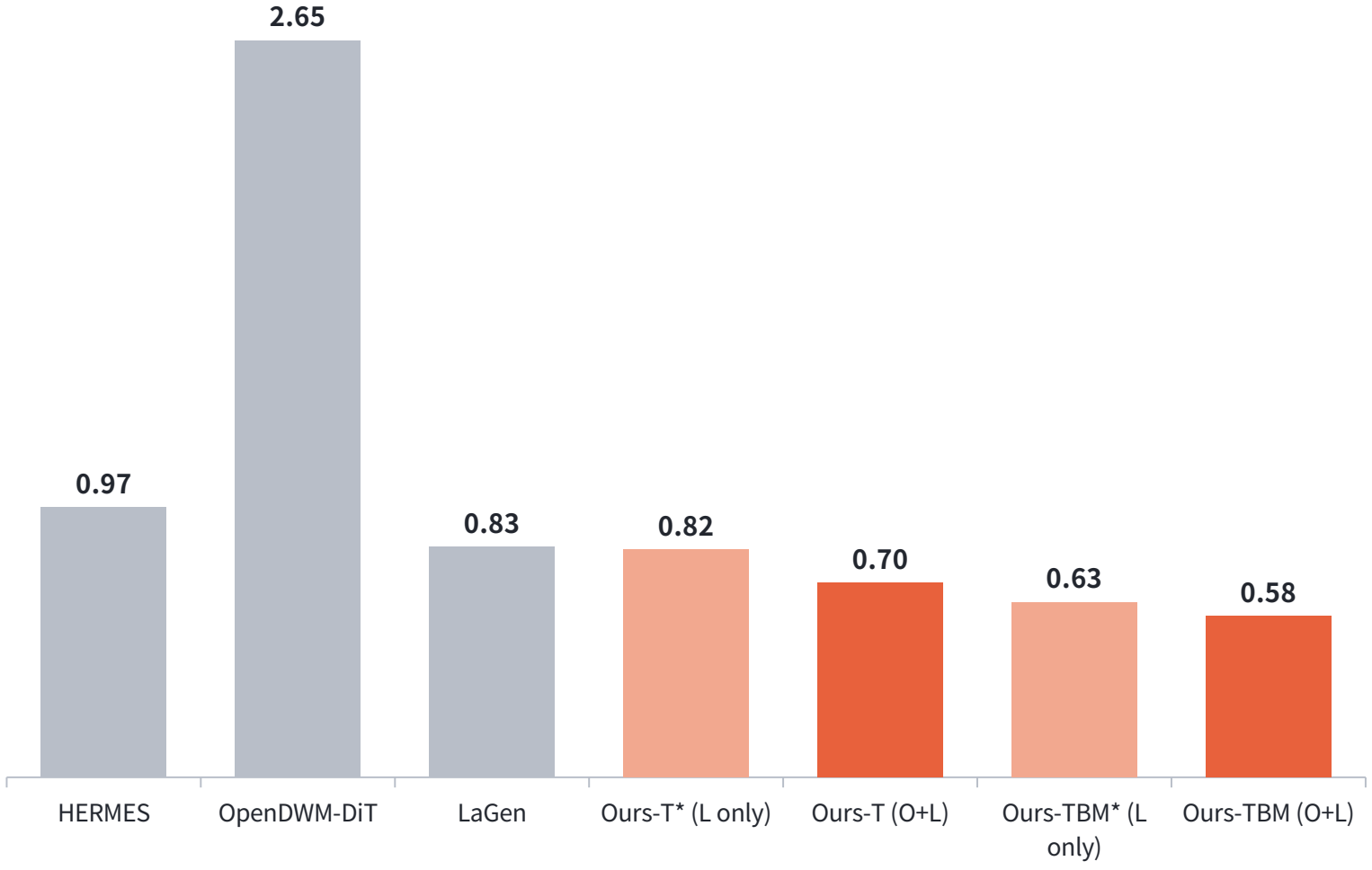
Camera occupancy input (STC)

IoU 31.69 vs I²-World-STC 28.77

Absolute scores drop markedly

Source: Table 1 (Occ3D-nuScenes val, avg. over 1/2/3 s)

LiDAR forecasting: CD falls to 0.58 m^2, two orders of magnitude faster



End-to-end speed FPS-E2E

OpenDWM-DiT	0.10
UniScene	0.19
HERMES	0.51
LO-World-T	26.26
LO-World-TBM	25.31

Distribution fidelity (FPD ↓)

Ours-TBM 4.68

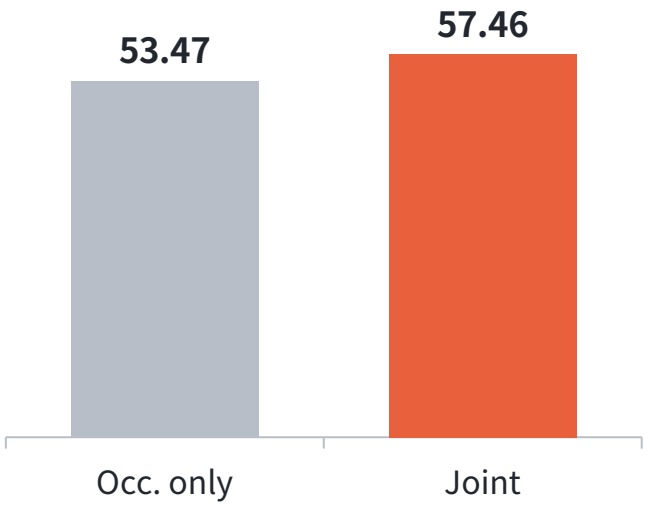
Ours-T 5.16 vs. LiDAR-only 8.84

OpenDWM-DiT 381.91

Source: Table 2 (nuScenes val, CD avg. over 1/2/3 s)

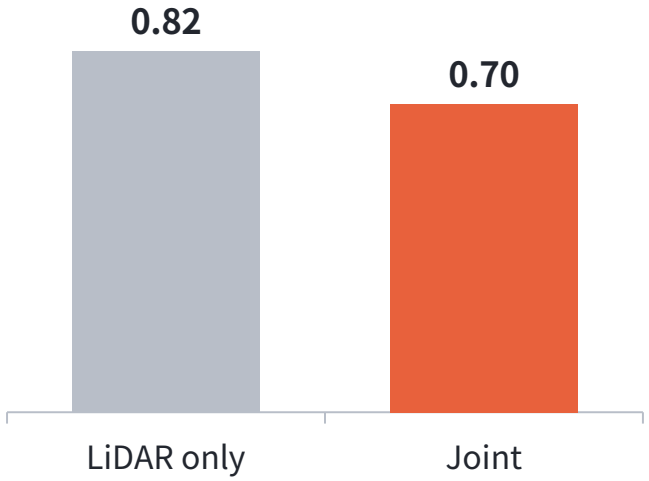
Key evidence: joint modeling improves both modalities at once

Occupancy Avg IoU (%) ↑



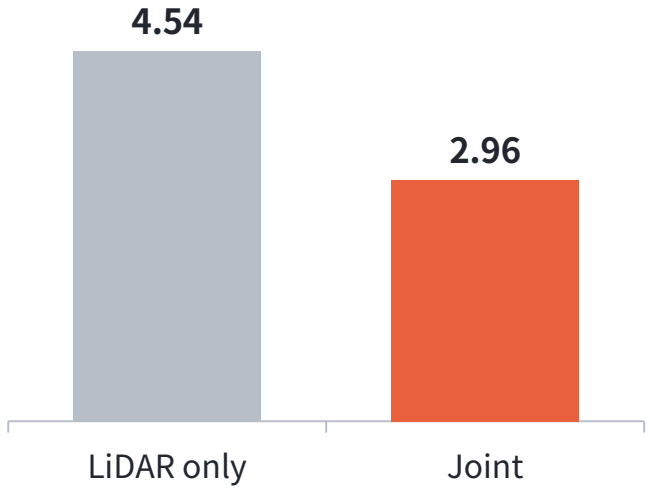
+3.99 pp

LiDAR Avg CD (m^2) ↓



−14.6%

B-JSD (×10^2) ↓



−34.8%

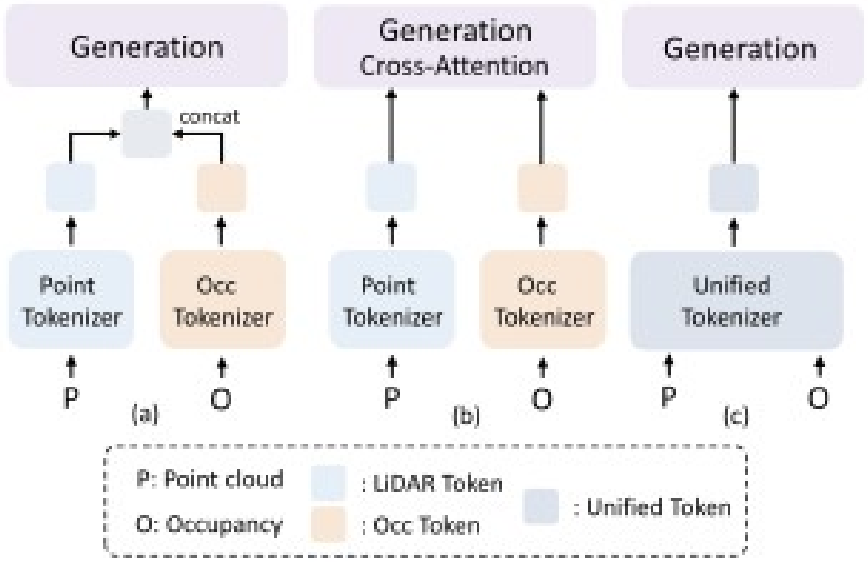
Robust to sampling steps (Table A5)

At 1 / 2 / 3 / 4 / 8 steps, the joint model beats its single-modality counterparts on mIoU, IoU and CD

Semantics aid geometry (Table 6)

Without occupancy semantic labels: CD 0.70 → 0.97, FPD 5.16 → 56.12

Shared latent beats concat & cross-attention, and runs faster



(a) Token concat (b) Cross-modal attention (c) Unified tokenizer (ours)

Setting	Avg IoU ↑	Avg CD ↓	FPD ↓	JC ↑	FPS-E2E ↑
(a) Separate latents + concat	54.08	0.84	9.01	0.52	17.76
(b) Separate latents + cross-attn	54.71	0.80	8.94	0.55	10.24
(c) Shared BEV latent (ours)	**57.46**	**0.70**	**5.16**	**0.77**	**26.26**
w/o hybrid repr. (plain voxels)	50.01	0.92	75.23	0.65	26.30
w/o hybrid repr. (high-res voxels)	55.87	0.75	32.59	0.67	6.72
w/o occ. semantic labels	–	0.97	56.12	0.60	26.27

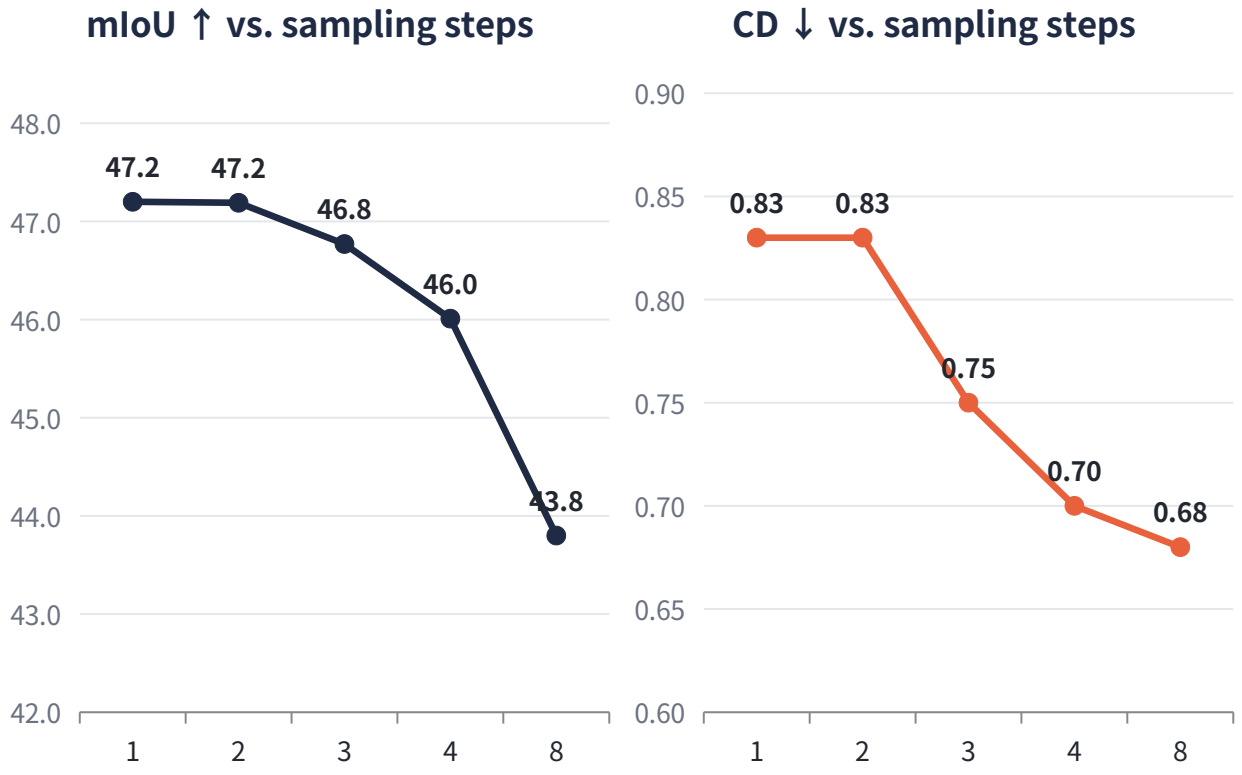
JC (joint consistency): harmonic mean of joint surface recall and occupancy precision; measures whether both modalities support the same geometry

Takeaway: moving fusion up to the tokenization stage is more accurate and cheaper than late fusion in the generator; the hybrid range-voxel representation gives the best accuracy-speed balance.

Context caching is key; 4-step sampling trades off accuracy and speed

Config (context, flow)	mIoU ↑	CD ↓	FPS-E2E ↑
Full model (8, 4)	**46.01**	**0.70**	**26.26**
w/o context block	37.19	1.58	34.34
w/o context block, 12-layer flow ‡	43.60	0.74	18.43
(4, 4)	42.81	0.90	31.71
(6, 4)	44.52	0.78	30.44
(8, 6)	46.36	0.68	24.12
(8, 2)	40.85	0.83	35.53

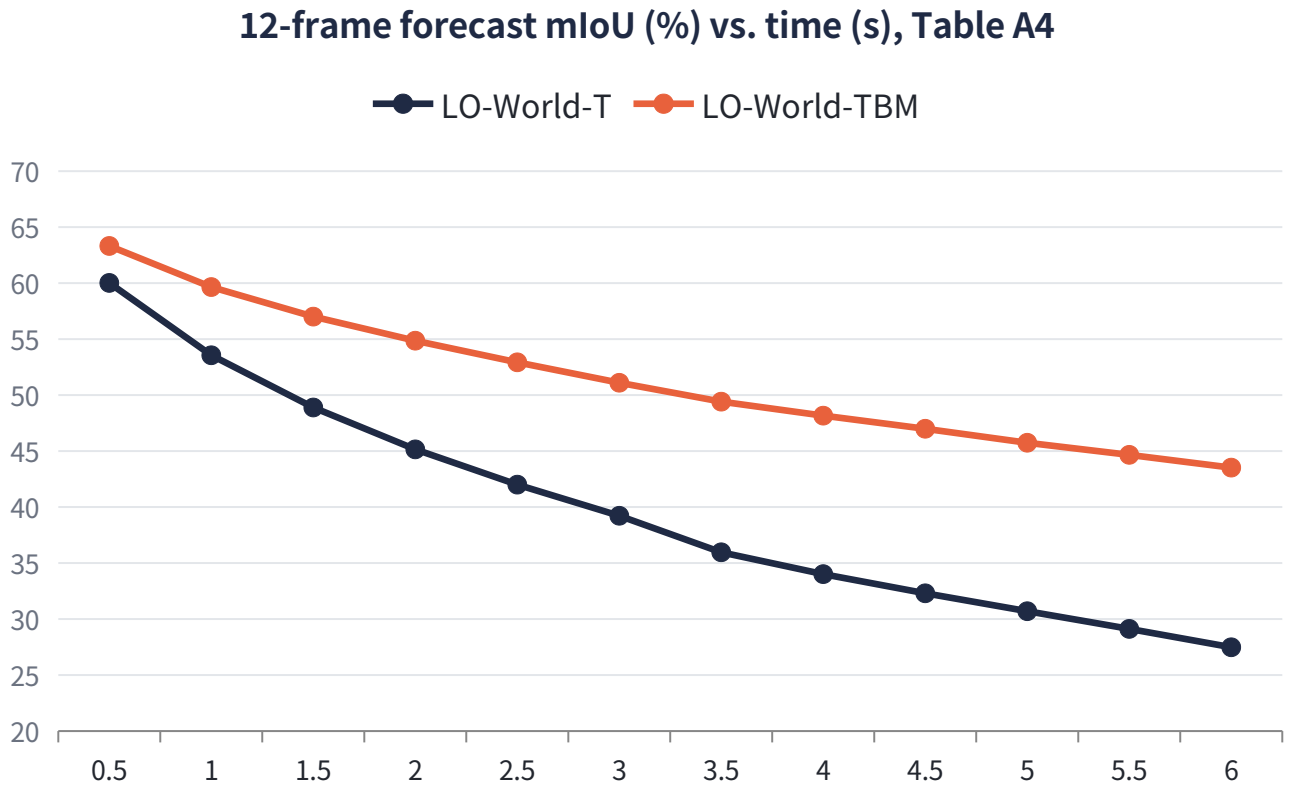
‡ Equal-depth control: 12-layer flow net, no context block. Decoupling context from the flow net is better in both accuracy and speed.



Trade-off: more steps improve geometry (CD) but hurt semantics (mIoU) and speed (1 step 40.06 → 8 steps 21.54 FPS-E2E), so 4 steps is the default.

Source: Tables 7, A5 (LO-World-T joint model)

Long horizons, cross-dataset, downstream video: gains largely hold



6 s mIoU： TBM 43.52 / T 27.48 / GenieDrive 23.66 / I^2-World 22.55

Caveat: the T variant's long-horizon LiDAR CD (1.49 at 4.0 s) trails LaGen (1.32)

Waymo cross-sensor (32-beam → 64-beam)

Zero-shot: mIoU 41.31 vs. I^2-World 36.38

Zero-shot CD@1s 0.91, below fine-tuned OpenDWM † (2.27)

Fine-tuned on 50 sequences (<7%): mIoU 46.00, CD@1s 0.51

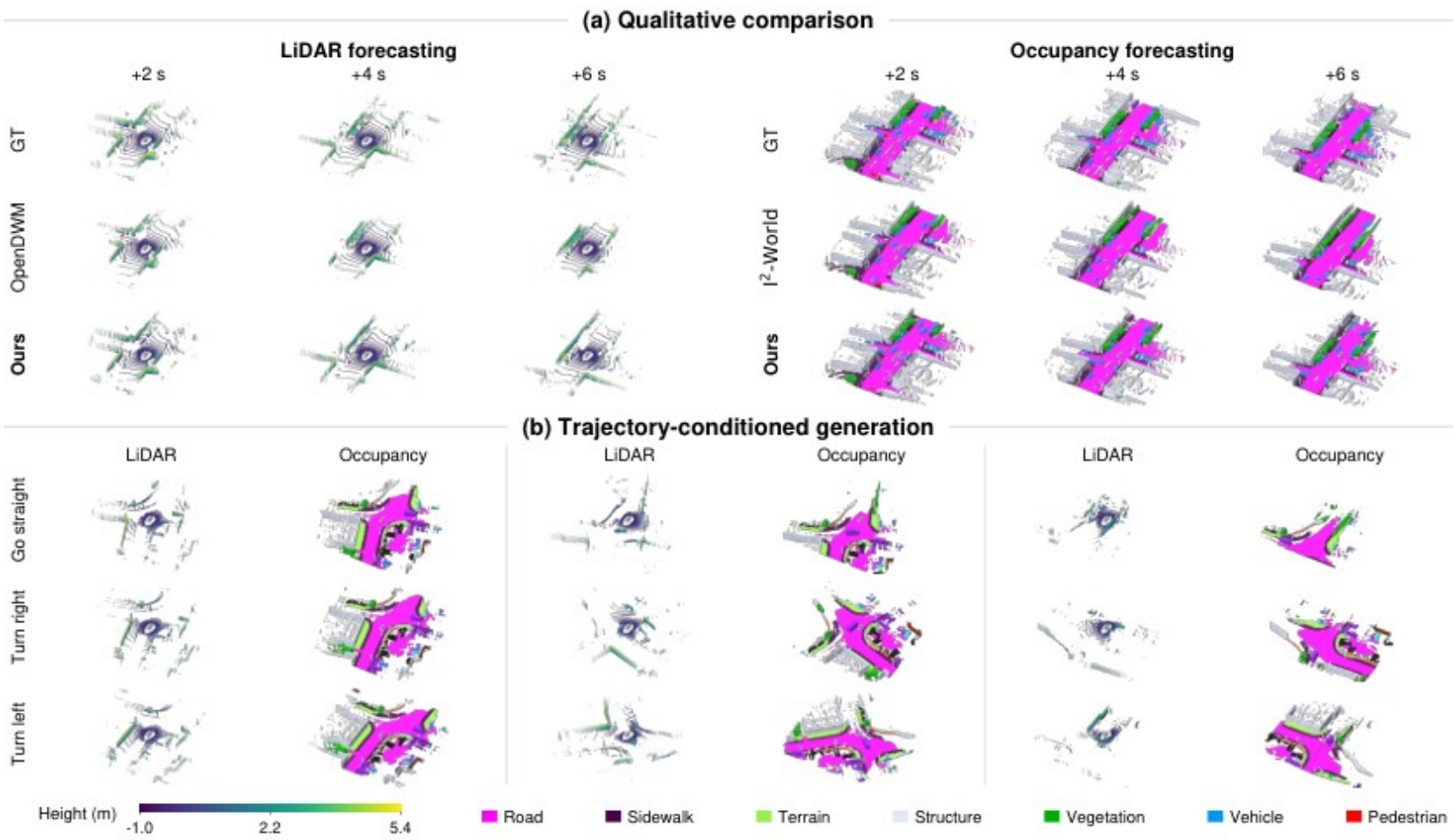
Downstream video generation (with GenieDrive)

FVD ↓: ours 60.41 vs. GenieDrive 92.78

mIoU 34.83 / mAP 27.15 (both above baseline)

Oracle (12 Hz GT occupancy) FVD 40.78: a gap remains

Qualitative results: more complete shapes, controllable by trajectory



(a) 2 / 4 / 6 s comparison

LiDAR: vs. OpenDWM, whose points become sparse and missing after 4 s

Occupancy: vs. I^2-World, our road layout is closer to GT

(b) Trajectory control

Same history, given straight / right-turn / left-turn trajectories

Occupancy and point-cloud forecasts change in sync, staying consistent across modalities

A sober look at the numbers: six caveats

① Unequal comparisons

+10.9% IoU compares O+L input with O-only GenieDrive; with same input it is 53.47 vs. 51.80 (≈ +3.2%)

② Relies on GT input

T/TBM take GT occupancy history; boxes and maps are external; camera occupancy input (STC) gives IoU of only 31.69

③ What FPS means

FPS-E2E counts future frames generated per second; one 6-frame rollout takes ≈ 228.5 ms (Table A6)

④ Diversity not evaluated

1-step sampling gives the best mIoU (47.20), closer to deterministic forecasting; no multi-sample diversity metric

⑤ Eval. protocol

mIoU excludes all classes with IoU=0 (Appendix B.2), unlike common protocols; JC is author-defined

⑥ Gen. & downstream

No reversing training data: occupancy incomplete, points sparse when reversing (Fig. A3); no perception or planning gains shown

Source: Tables 1/2/7, A6, Appendix B.2, Appendix D (Fig. A3)

Shared latent: simple multimodal world models

Early fusion	Merging occupancy and LiDAR into one BEV latent at tokenization is more accurate and faster than late concat or cross-attention
Mutual gains	In the same model, joint modeling gives IoU 53.47 → 57.46, CD 0.82 → 0.70; semantics also improve the point distribution
Efficient enough	Hybrid range-voxel encoding + context caching + 4-step sampling: ≈ 25 FPS end-to-end (RTX 4090)
Open questions	Relies on GT occupancy history; diversity unevaluated; best seen as a strongly conditioned, efficient forecaster or data generator

arXiv 2611.11912 • Code: github.com/lzzzzzm/LO-World